Big Data Project

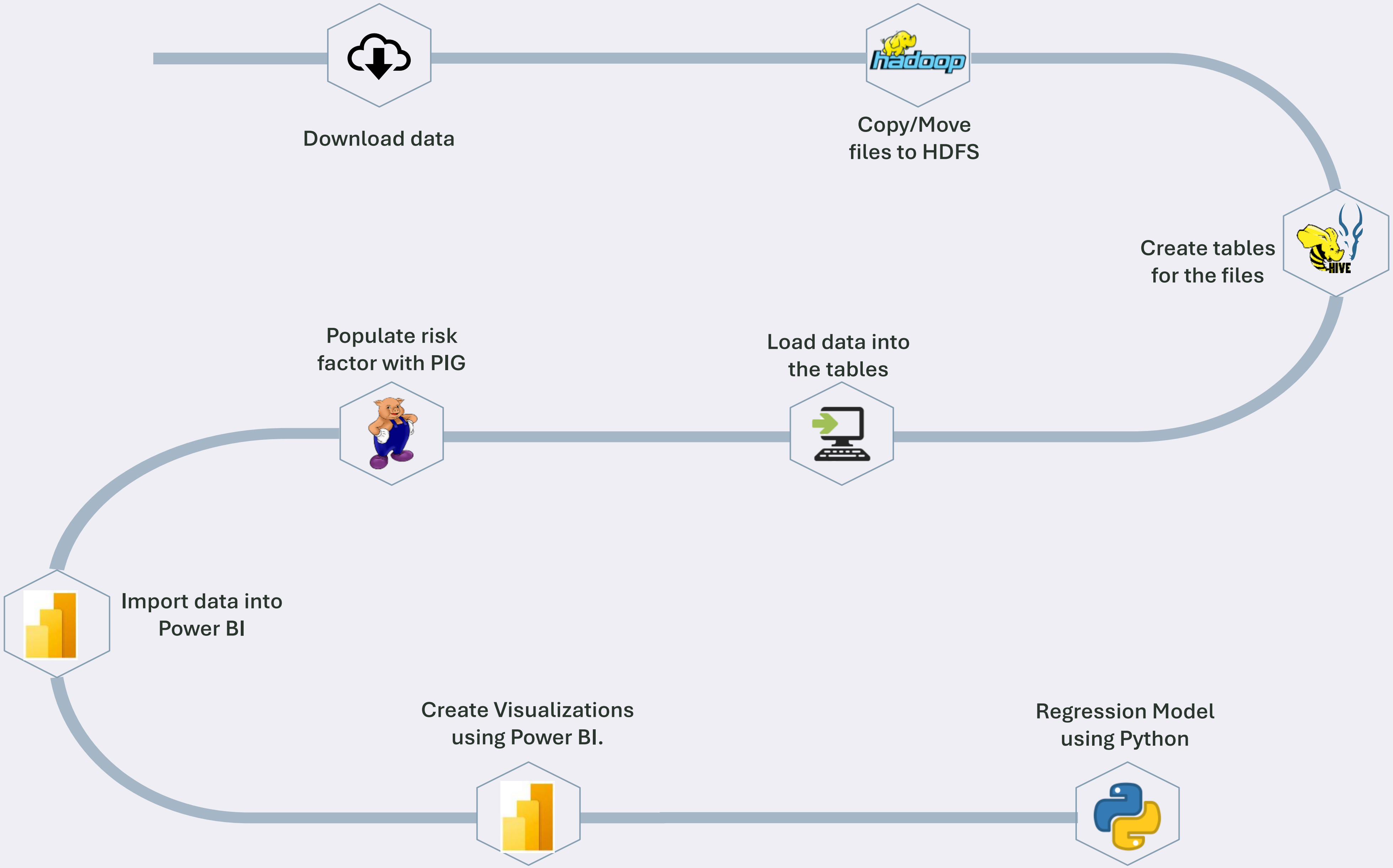
ANT Trucks Fleet Management

CONTENTS

OBJECTIVE

Utilize Big Data and Power BI analytics to optimize and improve fleet management by analyzing data. Use these insights for smarter route planning, reducing risk, and improving safety.

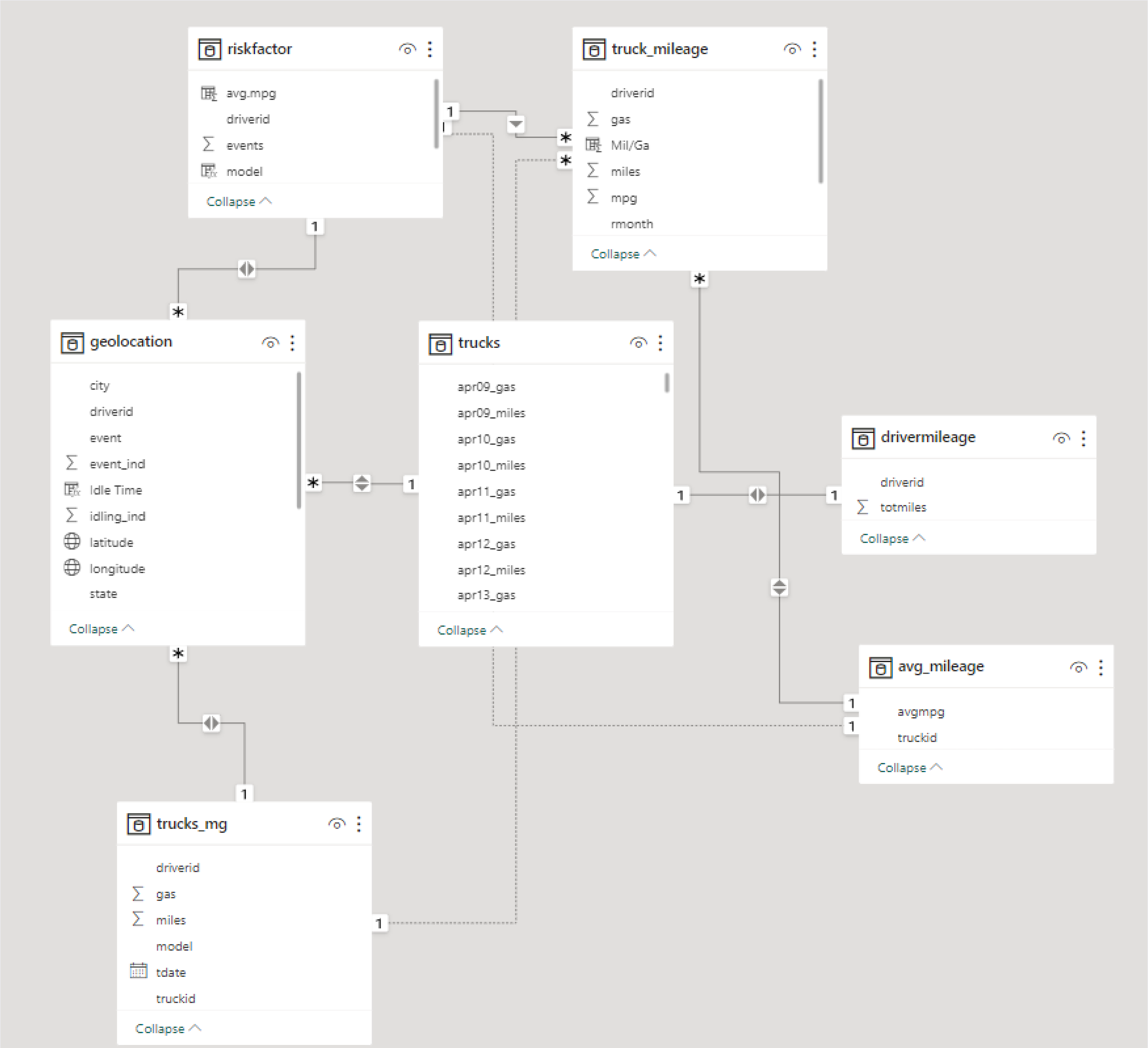
Workflow


Download data
Copy/Move files to HDFS
Create tables for the files
Load data into the tables
Populate risk factor with PIG
Import data into Power BI
Create Visualizations using Power BI.
Regression Model using Python

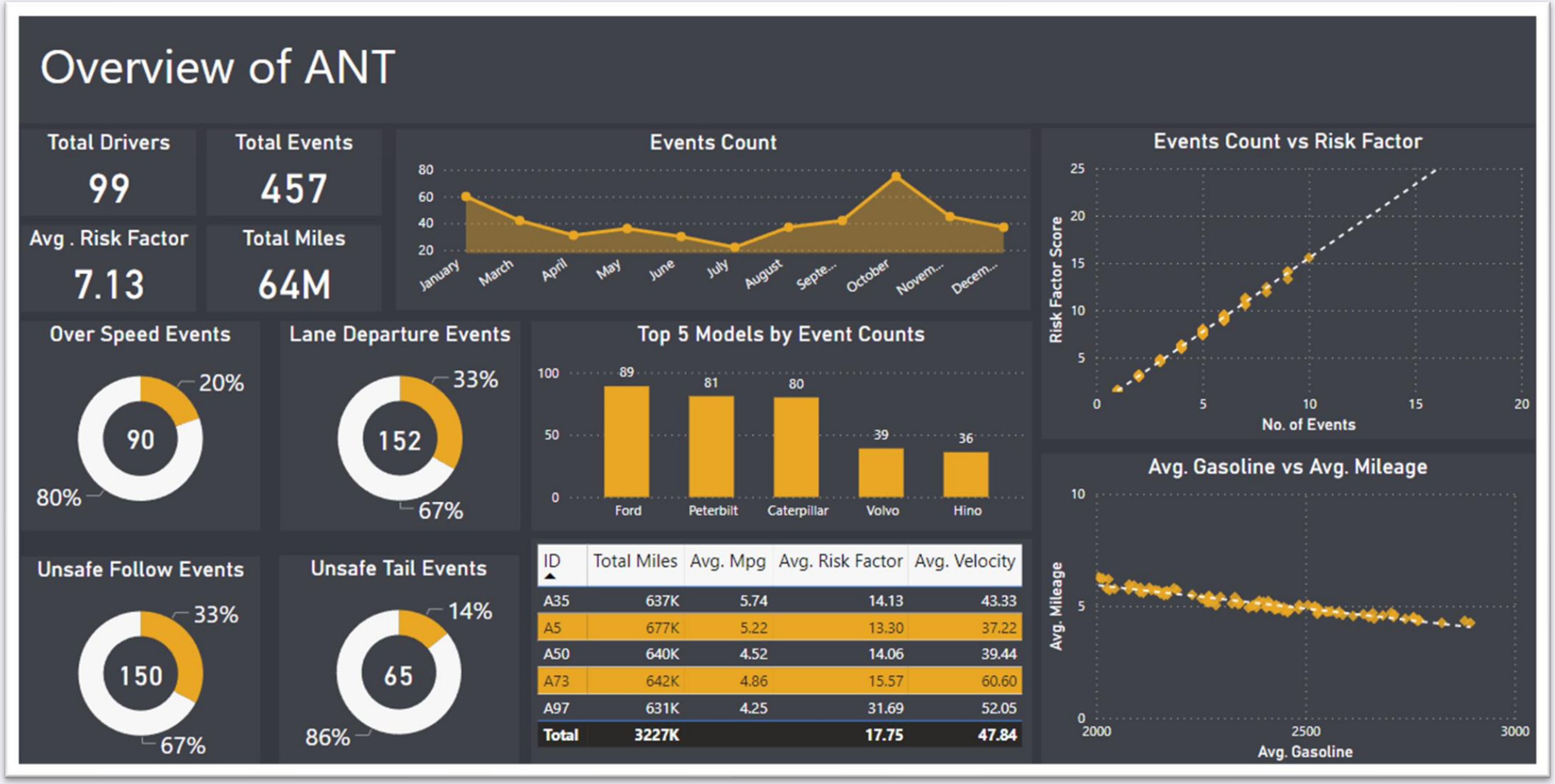
Visualizations

Analyze ANT's data for trends and patterns in driver behavior, safety incidents, model performance, and operational efficiency. Use the insights to facilitate data-driven decision-making and strategic planning.

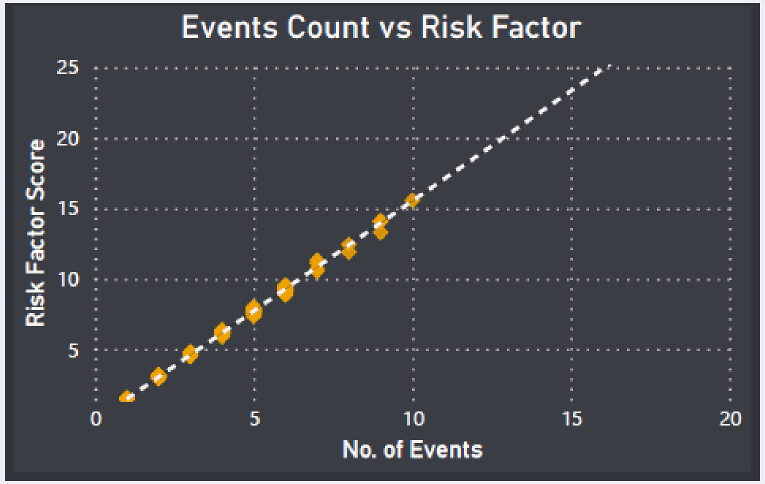
Relational Database



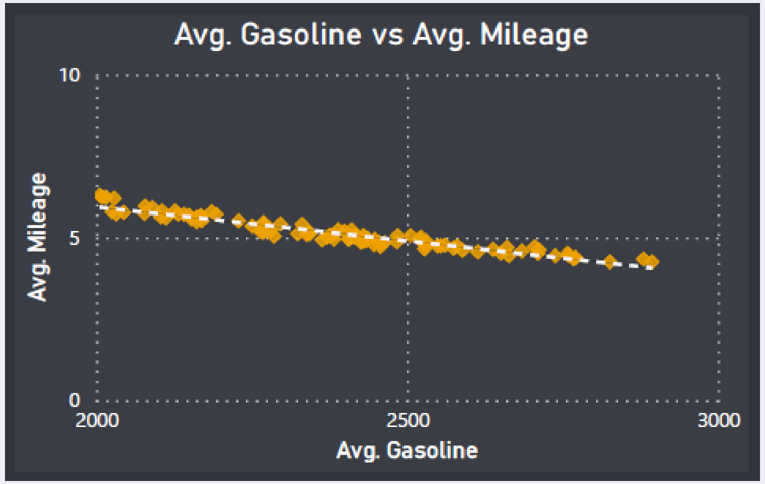
A Walkthrough of the ANT



1. Is there any correlation between the different factors?

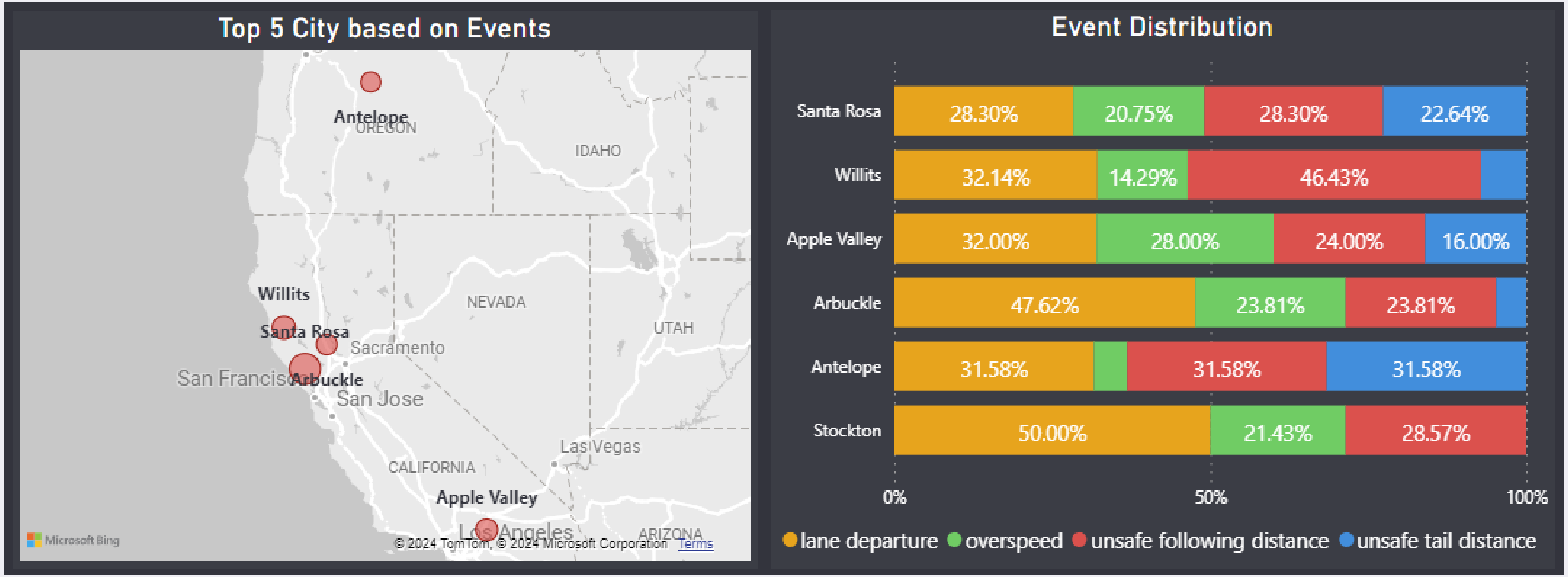


This relationship enables us to concentrate on cities and drivers with high event rates, which is critical for overall safety improvement.



This relationship illustrates the impact of fuel consumption on average mileage, enabling optimization of fuel use, reduced emissions, and improved operational performance.

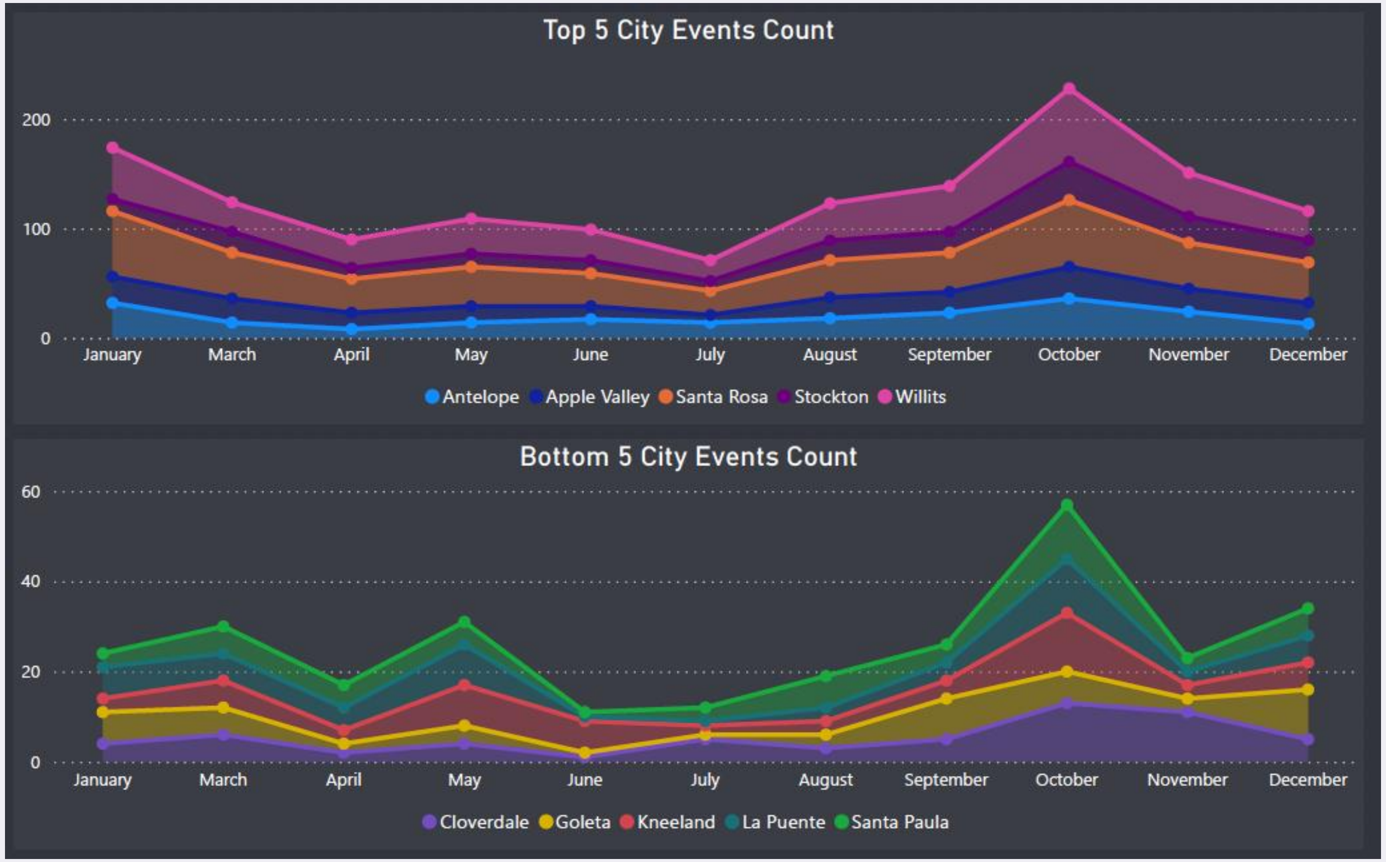
2. Which cities have more event occurrences?



Santa Rosa has the highest number of event occurrences, **53,** with an average risk factor of **8.32.**

Most of the events occur due to **lane departure** followed by **unsafe following distance**

3. What is the seasonality of the events?



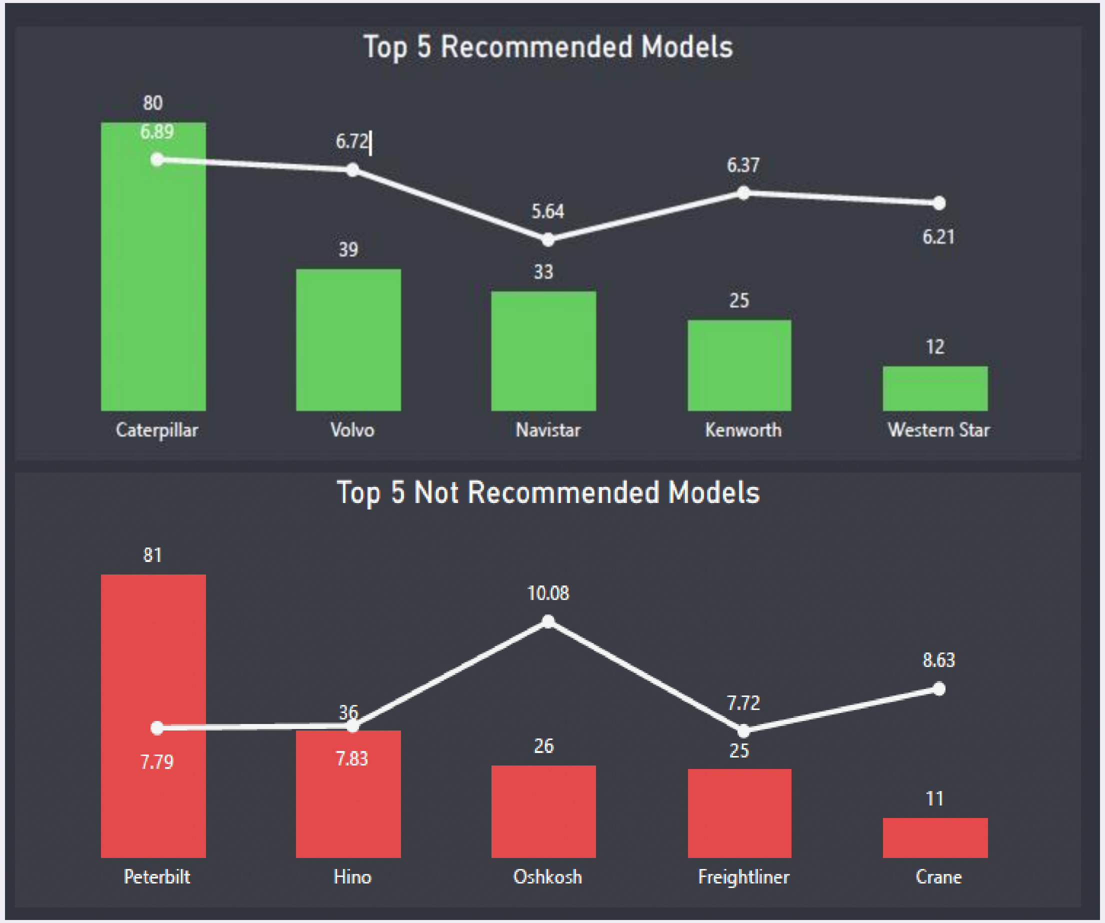
Months To Focus:

- January
- May
- October

The above months have event counts greater than the mean values.

There is a similarity of events occurrences with a slight variation in the seasonality of the bottom 5 cities. However, **October** had the highest number of event occurrences in both visuals.

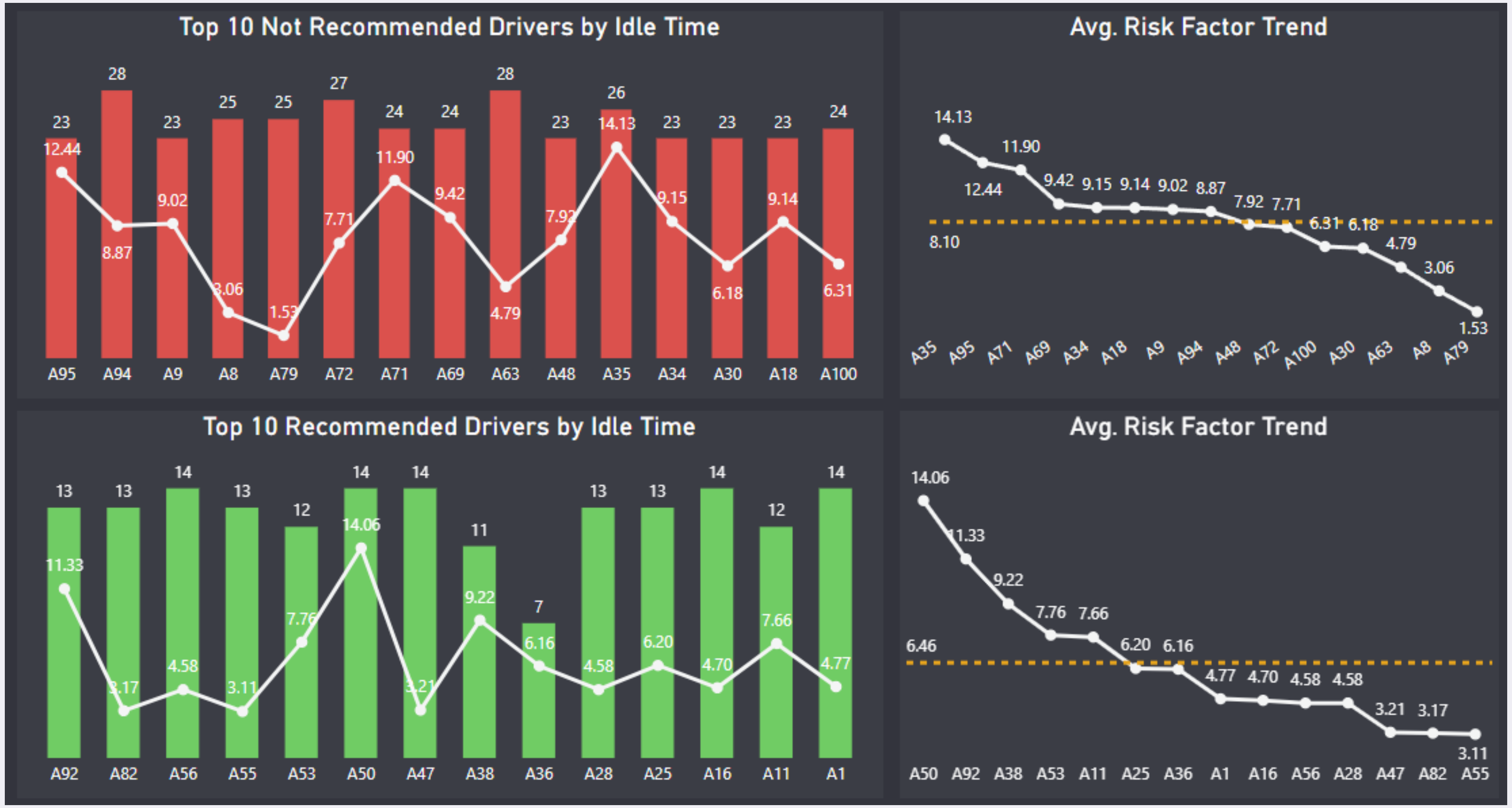
4. Which truck models are better for our fleet management?



Navistar is a better fit for our fleet due to lower risk factors. We must examine **Caterpillar** promptly, as it is close to breaching our threshold.

Oshkosh is considered to be the worst model for our fleet as the risk factor is very high with only 26 events. This trend is also seen in **Crane.**

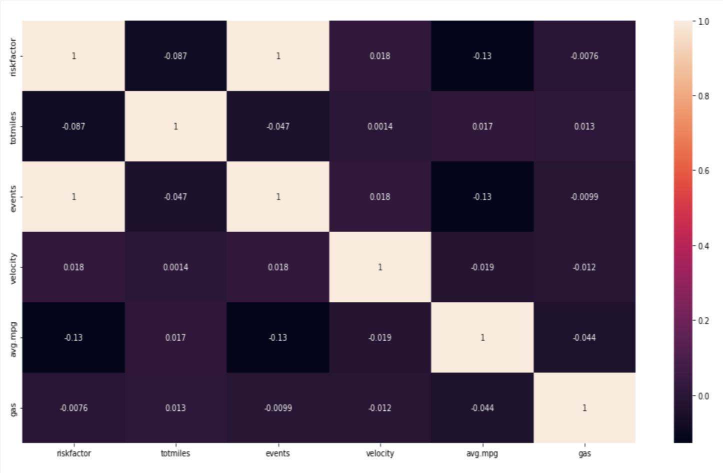
5. Is there a relation between Driver Idle Time and risk factor?



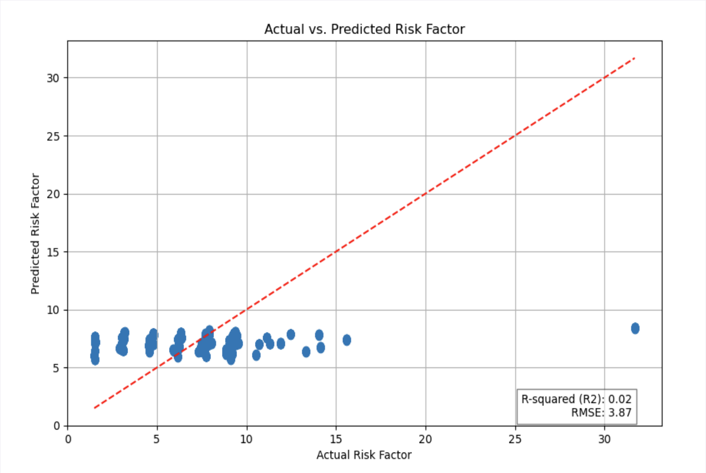
Examine why drivers with high idling events have high-risk factor scores to get a clear picture.

Promote the drivers for the trips having fewer idling events as they are associated with lower risk factor scores.

Regression Models using Python in Power BI



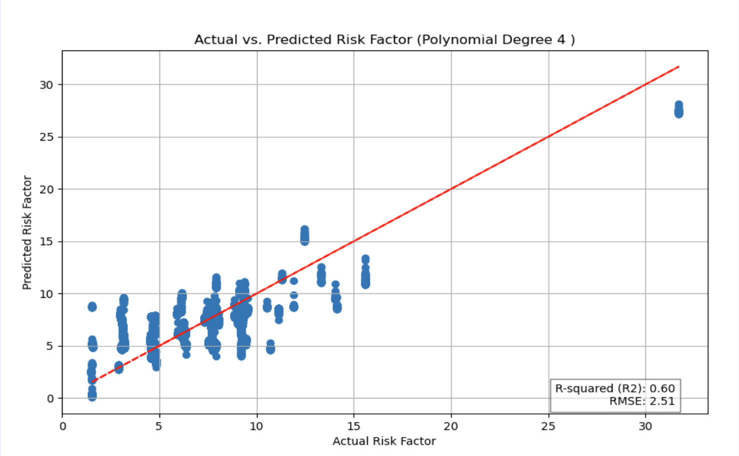
Correlation Heatmap



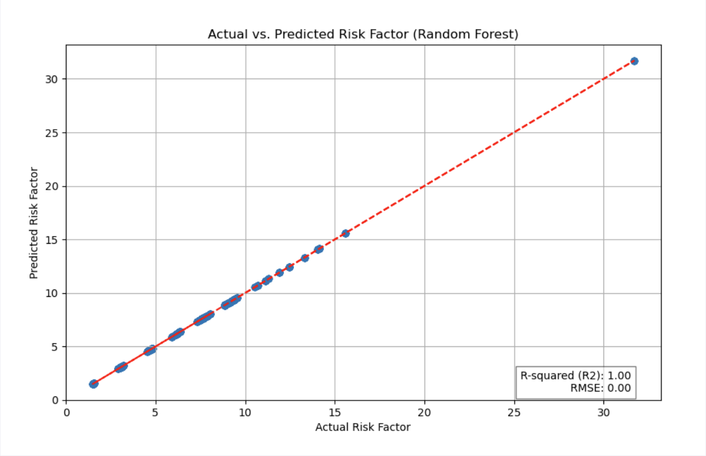
Multi-linear Regression Model

The following variables are used to create the regression models.

- Target: risk factor
- Predictors:
  - Gas
  - Total miles
  - Velocity
  - Avg. Mileage



Polynomial Regression Model (Degree 4)



Random Forest

Model Evaluation

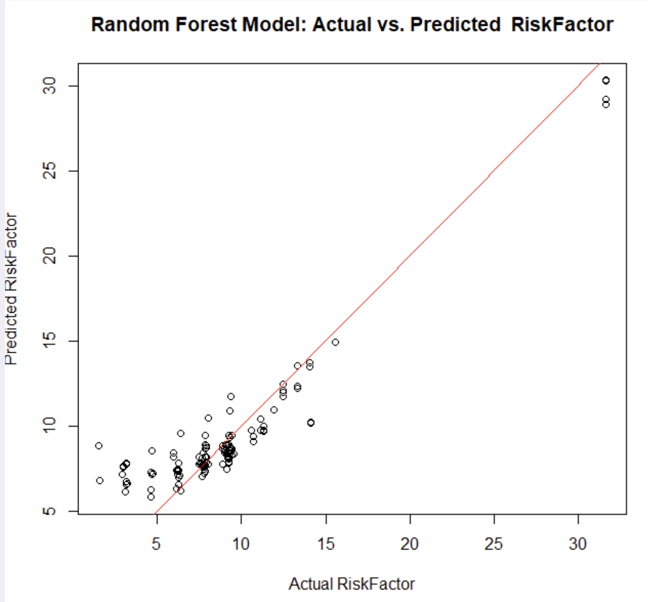
Best Model Fit: **Random Forest**

- R-Squared: 1.00
- R-MSE: 0.00

Regression Models using R-Studio

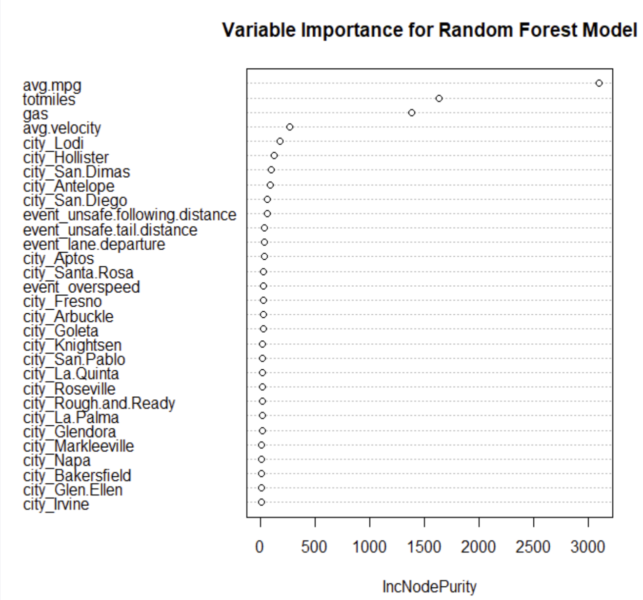
driverid	event	totmiles	city	avg.velocity	avg.mpg	avg.riskfactor	gas
A92	lane departure	618049	Apple Valley	74.00	5.79	11.33	2960
A92	lane departure	618049	San Diego	55.00	5.79	11.33	2960
A92	lane departure	618049	Willits	74.00	5.79	11.33	2960
A92	unsafe following distance	618049	Knights Landing	54.00	5.79	11.33	2960
A92	unsafe following distance	618049	San Pablo	66.00	5.79	11.33	2960
A92	unsafe following distance	618049	Santa Rosa	0.00	5.79	11.33	2960
A92	unsafe following distance	618049	Willits	23.00	5.79	11.33	2960
A26	lane departure	621790	Ukiah	45.00	5.04	4.82	2348
A26	overspeed	621790	San Francisco	94.00	5.04	4.82	2348
A26	unsafe following distance	621790	Willits	23.00	5.04	4.82	2348
A51	overspeed	622764	Modesto	86.00	5.71	8.03	2214
A51	unsafe following distance	622764	Antelope	31.00	5.71	8.03	2214
A51	unsafe following distance	622764	Lodi	24.00	5.71	8.03	2214
A51	unsafe tail distance	622764	La Palma	53.00	5.71	8.03	2214
A47	overspeed	623836	Santa Rosa	88.00	4.72	3.21	2672
A47	unsafe tail distance	623836	Santa Rosa	15.00	4.72	3.21	2672
A62	lane departure	626253	Antelope	45.00	5.77	4.79	2061
Total		**64176456**		**46.19**			

The table summary was created using Power BI, then exported as CSV to build regression models in R-Studio.



Random Forest

R-Squared: 0.881
R-MSE: 1.833



Variable Importance

1. Avg. Mileage
2. Total Miles
3. Gas
4. Avg. Velocity

Challenges

- Implementing regression models in Power BI using Python script visual.
- Finding the correlation between various parameters.
- Configuring ODBC settings to connect the Hadoop server from Power BI.

Conclusion

- Focus on cities with higher event occurrences, such as **Santa Rosa**, and implement targeted safety measures.
- Investigate the events seasonality in months such as **January, May, and October** to understand underlying factors.
- Assess truck models' performance and prioritize **Navistar** while addressing issues with models like **Oshkosh and Crane**.
- Analyze **Driver Idle Time's correlation with risk factor** scores to optimize driver behavior and safety protocols.
- Utilize data-driven strategies for safety, operational efficiency, and emission reduction based on the analysis findings.

Thank you!